



PROJECT DURATION

45 Days



CURRENT SPRINT

SPRINT 6

Intelligence, ML, Reports,
API & QA
(Days 36-45)

CURRENT MODULES

Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)

VERSION 1.0

July 15, 2026

SPRINT 6 — INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36-45)

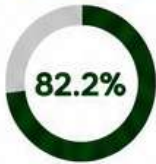
DAY 37 COMPLETED



JULY 15, 2026 (Today)

Days 36 & 37 Completed Successfully!

TOTAL PROJECT PROGRESS



37 / 45 Days Completed

Completed 37
In Progress 0
Pending 8



SPRINT 6 OBJECTIVE

Perform advanced statistical analysis & clustering, build REST API, execute test suite & QA, generate automated reports and deliver production-ready release.

Statistical Analysis & Clustering

REST API Development

Testing, QA & Documentation

Automated Reporting (PDF)

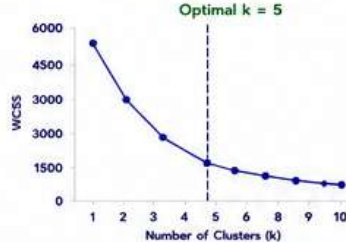
DAY 36 COMPLETED

July 15, 2026 – Statistical Analysis & Clustering (Module 10 – Part 1)

10.1 KMEANS CLUSTERING (5 CLUSTERS)

- Features: ROE, D/E, Revenue CAGR, FCF CAGR, OPM
- StandardScaler + KMeans(n=5)
- Elbow method used to validate k
- 5 clusters generated

ELBOW METHOD (WCSS)



10.2 CLUSTER PROFILING (MEAN VALUES)

Cluster	Cluster Name	No. of Companies	ROE (%)	D/E (x)	Revenue CAGR (%)	FCF CAGR (%)	OPM (%)
0	High-Quality Growth	23	24.8	0.42	16.2	18.7	19.6
1	Defensive Dividend	19	15.1	0.35	6.3	7.2	17.1
2	Value Cyclicals	21	13.2	0.98	8.9	9.1	14.2
3	Emerging Growth	17	18.3	0.74	22.1	20.5	15.8
4	Distressed / Turnaround	12	-1.8	2.41	-2.6	-3.1	5.6

CLUSTER DISTRIBUTION



High-Quality Growth 23 (25.0%)
Defensive Dividend 19 (20.7%)
Value Cyclicals 21 (22.8%)
Emerging Growth 17 (18.5%)
Distressed / Turnaround 12 (13.0%)

OUTPUT FILE

cluster_labels.csv
(company_id, cluster_id,
cluster_name,
distance_from_centroid)

Total Rows: 92

KEY DELIVERABLE – DAY 36

- KMeans clustering completed with 5 distinct clusters
- Cluster profiles generated and labeled

DAY 37 COMPLETED

July 15, 2026 – Statistical Analysis & Clustering (Module 10 – Part 2)

10.3 CORRELATION MATRIX (10 KPIs)

	ROE	ROCE	OPM	D/E	P/E	CFP/PAT	FCF/PAT	Rev CAGR
ROE	1.00	0.84	0.65	-0.41	-0.32	0.37	0.46	0.55
ROCE	0.84	1.00	0.67	-0.47	-0.36	0.42	0.50	0.67
OPM	0.65	0.67	1.00	-0.25	-0.15	0.36	0.36	0.48
D/E	-0.41	-0.47	-0.25	1.00	0.32	-0.42	-0.21	-0.33
P/E	-0.32	-0.36	-0.15	0.32	1.00	-0.16	-0.16	-0.11
CFP/PAT	0.37	0.42	0.36	-0.42	-0.16	1.00	0.38	0.31
FCF/PAT	0.46	0.50	0.36	-0.21	-0.16	0.38	1.00	0.56
Rev CAGR	0.55	0.67	0.48	-0.33	-0.11	0.31	0.56	1.00

Output File: correlation_heatmap.png

10.5 PORTFOLIO STATISTICS (92 COMPANIES)

Metric	P10	P25	P50	P75	P90	Mean	Std
ROE (%)	7.2	12.8	18.6	24.7	31.4	19.1	7.8
D/E (x)	0.15	0.45	0.85	1.35	1.95	0.92	0.63
P/E (x)	11.6	18.2	24.8	31.7	42.6	27.3	8.7
P/B (x)	1.2	1.7	2.4	3.5	5.1	2.8	1.2
OPM (%)	8.1	12.7	16.2	20.1	26.3	16.6	5.3
Revenue CAGR (%)	3.8	7.6	11.9	16.2	21.8	12.8	5.6
FCF CAGR (%)	2.1	6.3	10.4	14.9	20.2	11.1	5.9
PAT CAGR (%)	2.5	6.8	10.9	15.1	20.8	11.6	6.1

Output File: portfolio_stats.csv

KEY DELIVERABLE – DAY 37

- Correlation matrix, outlier detection & portfolio statistics completed
- All statistical analysis outputs generated successfully

SPRINT 6 PROGRESS (DAYS 36-45)



Completed Days : 2 / 10
In Progress Days : 0 / 10
Pending Days : 8 / 10
Overall Progress : 13.3%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



Completed Days : 37 / 45
In Progress Days : 0 / 45
Pending Days : 8 / 45
Overall Progress : 82.2%

Next Milestone:

End of Day 42
Sprint 6 Review
(API Demo & Test
Report Review)

SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1-7) : 100%
Sprint 2 (Days 8-14) : 100%
Sprint 3 (Days 15-21) : 100%
Sprint 4 (Days 22-28) : 100%
Sprint 5 (Days 29-35) : 100%
Sprint 6 (Days 36-45) : 13.3%



KEY DELIVERABLES GENERATED TILL TODAY (DAYS 36 & 37):

cluster_labels.csv
(Cluster Labels)correlation_heatmap.png
(10 KPIs Correlation)outlier_report.csv
(Outlier Report)portfolio_stats.csv
(Portfolio Statistics)DATA-DRIVEN INSIGHTS
Better DecisionsFINANCIAL EXCELLENCE
Stronger PerformanceINTELLIGENT INVESTING
Higher ReturnsBUILT FOR SCALE
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INTELLIGENCE PLATFORM

PROJECT DURATION
45 DaysCURRENT SPRINT
SPRINT 6
Intelligence, ML, Reports,
API & QA
(Days 36–45)CURRENT MODULES
Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)VERSION 1.0
July 16, 2026

SPRINT 6 — INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36–45)

DAY 39 COMPLETED

JULY 16, 2026 (Today)

TOTAL PROJECT PROGRESS



38 / 45 Days Completed

Completed 39
In Progress 0
Pending 6



DAY 38 COMPLETED

July 16, 2026 — REST API Module (Part 1) — Setup & Core Endpoints (11.1–11.5)

SERVER SETUP & SCAFFOLD

- FastAPI app initialized in src/api/main.py
- SQLite connection function created
- CORS middleware (allow all origins)
- Request logging middleware (method, path, response time)
- Routers/ directory created (one file per module)
- All routers imported with prefix /api/v1
- GET /api/v1/health endpoint implemented
- Uvicorn server running on port 8000
- OpenAPI docs available at /docs

Health Check Response

```
{
  "status": "ok",
  "version": "1.0.0",
  "uptime_seconds": 1234,
  "db_row_counts": {
    "companies": 92,
    "sectors": 11,
    "profitandloss": 5520,
    "balancesheet": 5550,
    "cashflow": 5530,
    "ratios": 12880,
    "peer_cons": 2208,
    "analysts_paired": 2760,
    "cashflow_intelligence": 92,
    "cluster_labels": 92
  }
}
```

CORE COMPANY & FINANCIAL ENDPOINTS

Endpoint	Description
11.1 /api/v1/companies	List all 92 companies with id, name, sector. Supports ?sector= filter.
11.2 GET /api/v1/companies/{ticker}	Full company profile: KPIs, pros/cons, sector, description.
11.3 GET /api/v1/companies/{ticker}/pl	P&L history with ?from_year= & ?to_year=
11.4 GET /api/v1/companies/{ticker}/bs	Balance sheet history with ?from_year= & ?to_year=
11.5 GET /api/v1/companies/{ticker}/cf	Cash flow history with ?from_year= & ?to_year=

ROUTERS CREATED

companies.py	(Company Endpoints)
screener.py	(Screener Endpoints)
sectors.py	(Sectors Endpoints)
peers.py	(Peers & Comparison)
valuation.py	(Valuation Endpoints)
portfolio.py	(Portfolio Statistics)
documents.py	(Documents & Reports)
health.py	(Health & System Info)

OUTPUTS GENERATED

openapi.json (Auto-generated OpenAPI 3.0 spec)	postman_collection.json (Exported Postman Collection)
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DAY 39 COMPLETED

July 16, 2026 — REST API Module (Part 2) — Company Data & Key Endpoints (11.6–11.10)

COMPANY DATA & KPI ENDPOINTS

Endpoint	Description
GET /api/v1/companies	List of all 92 companies with key fields (id, name, sector, sub_sector, roe_pct, roce_pct). Supports filters: sector, market_cap_category, search.
GET /api/v1/companies/{ticker}	Full company profile with latest year KPIs and sector data. Returns 404 if not found.
GET /api/v1/companies/{ticker}/pl	P&L history array with from_year & to_year.
GET /api/v1/companies/{ticker}/bs	Balance sheet history array with from_year & to_year.
GET /api/v1/companies/{ticker}/cashflow	Cash flow history array with from_year & to_year.
GET /api/v1/companies/{ticker}/ratios	All 14 pre-computed KPIs per year for the company.
GET /api/v1/companies/{ticker}/tearsheet	Returns pre-generated tearsheet PDF (binary file download).
GET /api/v1/screener	Screener endpoint with query params ?min_roe=, ?max_debt_to_equity=, ?min_cagr=, etc.) returns ranked list.
GET /api/v1/sectors	List all 11 sectors with company count and median KPIs.
GET /api/v1/sectors/{sector}/companies	All companies in a sector with KPI summary.

OUTPUTS GENERATED

openapi.json (Auto-exported OpenAPI 3.0 spec) – Updated with all endpoints	postman_collection.json (Updated Postman Collection with all endpoints)
--	---

SPRINT 6 PROGRESS (DAYS 36–45)



Completed Days : 4 / 15
In Progress Days : 0 / 15
Pending Days : 11 / 15
Overall Progress : 26.7%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



Completed Days : 39 / 45
In Progress Days : 0 / 45
Pending Days : 6 / 45
Overall Progress : 84.4%

Next Milestone:
End of Day 42
Sprint 6 Review
(API Demo & Test
Report Review)

SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1–7) : 100%
Sprint 2 (Days 8–14) : 100%
Sprint 3 (Days 15–21) : 100%
Sprint 4 (Days 22–28) : 100%
Sprint 5 (Days 29–35) : 100%
Sprint 6 (Days 36–45) : 26.7%



KEY DELIVERABLES GENERATED TILL TODAY (DAYS 36 & 37):

cluster_labels.csv (Cluster Labels)	correlation_heatmap.png (10 KPIs Correlation)	outlier_report.csv (Outlier Report)	portfolio_stats.csv (Portfolio Statistics)	openapi.json (OpenAPI Spec)	postman_collection.json (Postman Collection)
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DATA-DRIVEN INSIGHTS
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PROJECT DURATION
45 Days



CURRENT SPRINT
SPRINT 6
Intelligence, ML, Reports,
API & QA
(Days 36-45)



CURRENT MODULES
Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)



VERSION 1.0
July 16, 2026



SPRINT 6 — INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36-45)

DAY 41 COMPLETED

JULY 17, 2026 (Today)

Days 36 to 41 Completed Successfully!

TOTAL PROJECT PROGRESS



41 / 45 Days Completed

Completed 41
In Progress 0
Pending 4



SPRINT 6 OBJECTIVE

Perform advanced statistical analysis & clustering, build REST API, execute test suite & QA, generate automated reports and deliver production-ready release.

Statistical Analysis & Clustering
REST API Development

Testing, QA & Documentation
Automated Reporting (PDF)

DAY 40 COMPLETED

July 17, 2026 — REST API Module (Part 3) — Screener, Sector, Peer & Remaining (11.11-11.16)

API ENDPOINTS IMPLEMENTED (11.11-11.16)

Endpoint	Description
GET /api/v1/screener	Screener with filters: min_roe, max_de, min_fcf, sector, min_rev_cagr_5yr, min_pat_cagr_5yr, max_pe. Returns ranked list.
GET /api/v1/sectors	All 11 sectors with company_count, median_roe, median_pe, median_de.
GET /api/v1/sectors/{sector}/companies	All companies in a sector with latest year KPIs.
GET /api/v1/peers/{group_name}	Companies in peer group with percentile rank for 10 metrics.
GET /api/v1/companies/{ticker}/peers/compar	Radar data: company vs peer group average + benchmark company (8 axis metrics).
GET /api/v1/market-cap/{ticker}	Historical valuation multiples (P/E, P/B, EV/EBITDA) 2019-2024.
GET /api/v1/portfolio/stats	P10-P90 percentile table for 10 core KPIs across all 92 companies.
GET /api/v1/companies/{ticker}/documents	Annual report links with is_url_valid boolean flag.
GET /api/v1/health	Server health check: DB row counts, uptime, version.

EXPORTS & DOCUMENTATION



openapi.json
Auto-exported OpenAPI 3.0 spec.
Import into Postman.



postman_collection.json
Postman collection exported
with all 16 endpoints.

Total API Endpoints Implemented: 16 / 16

DAY 41 COMPLETED

July 17, 2026 — Testing & QA Module (Part 1) — ETL & KPI Unit Tests (12.1-12.2)

TEST SUITE CREATED & EXECUTED

Test Suite	Tests	Coverage Area	Status
tests/etl/test_normalise.py	20	normalize_year() unit tests	✓
tests/etl/test_loader.py	10	Loader row counts, column names validation	✓
tests/kpi/test_ratios.py	20	ROE, D/E edge cases, ICR (interest=0), CAGR turnaround, decline-to-loss, CFO quality score	✓
tests/dq/test_rules.py	14	14 DQ rules - rule_id & severity validation	✓

TEST SUMMARY

Total Tests Executed

64

Passed

64

Failed

0

Pass Rate

100%

COVERAGE AREAS VALIDATED

12.1 ETL TESTS (45 TESTS)

- year_normaliser (20 cases)
- ticker_normaliser (15 cases)
- deduplication (5 cases)
- load_audit completeness (5 cases)

ALL 45 TESTS PASSED

12.2 KPI FORMULA TESTS (20 TESTS)

- ROE with positive equity
- ROE with negative equity (None)
- D/E for debt-free (0)
- ICR for interest=0 (None)
- CAGR turnaround flag
- OPM cross-check

ALL 20 TESTS PASSED

All unit tests passed successfully. No failures.

SPRINT 6 PROGRESS (DAYS 36-45)



Completed Tasks : 6 / 10
In Progress Tasks : 0 / 10
Pending Tasks : 4 / 10
Overall Sprint Progress : 60.0%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



Completed Days : 41 / 45
In Progress Days : 0 / 45
Pending Days : 4 / 45
Overall Progress : 91.1%

Next Milestone:

End of Day 42
Sprint 6 Review
(API Demo & Test
Report Review)



SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1-7) : 100%
Sprint 2 (Days 8-14) : 100%
Sprint 3 (Days 15-21) : 100%
Sprint 4 (Days 22-28) : 100%
Sprint 5 (Days 29-35) : 100%
Sprint 6 (Days 36-45) : 60.0%



KEY DELIVERABLES GENERATED TILL TODAY (DAYS 36 & 37):

- cluster_labels.csv (Cluster Labels)
- correlation_heatmap.png (10 KPIs Correlation)
- outlier_report.csv (Outlier Report)
- portfolio_stats.csv (Portfolio Statistics)
- openapi.json (OpenAPI Spec)
- postman_collection.json (Postman Collection)
- pytest_report.html (Test Report - WIP)



DATA-DRIVEN INSIGHTS
Better Decisions



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PROJECT DURATION

45 Days



CURRENT SPRINT

SPRINT 6

Intelligence, ML, Reports,
API & QA
(Days 36-45)



CURRENT MODULES

Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)



VERSION 1.0

July 20, 2026



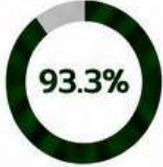
SPRINT 6 — INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36-45)

DAY 42 COMPLETED



JULY 20, 2026 (Today)

TOTAL PROJECT PROGRESS



42 / 45 Days Completed

Completed 42
In Progress 0
Pending 3



SPRINT 6 OBJECTIVE

Perform advanced statistical analysis & clustering, build REST API, execute test suite & QA, generate automated reports and deliver production-ready release.

Statistical Analysis & Clustering
REST API Development

Testing, QA & Documentation
Automated Reporting (PDF)

DAY 42 COMPLETED

July 20, 2026 — Testing & QA Module (Part 2) — API Tests & Integration (12.3-12.5)



API TESTS & INTEGRATION

- API tests written for: health endpoint, companies, screener, and sectors
- Generated pytest HTML report: `pytest tests/ --html=reports/pytest_report.html` — shows ≥ 60 tests with 0 failures
- Integration test: Verified dashboard data loads correctly from the API
- Verified: Screener results in Streamlit app match API screener endpoint results



12.3 DQ RULE TESTS

- Each of 14 DQ rules triggered on crafted violation records
- Severity levels validated correctly
- Edge cases and null values tested
- All rules triggered successfully

14 / 14 Rules Triggered Correctly



12.4 API TESTS

- /health returns 200
- /companies returns 92 records
- /screener with valid filters returns ranked list
- Invalid ticker returns 404
- Sectors & other endpoints verified

All 10 API Tests Passed

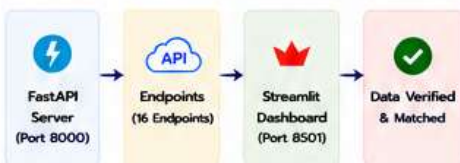


12.5 REGRESSION TESTS

- Full test suite executed (60+ tests)
- 0 critical bugs
- All previously-passing tests intact
- Coverage maintained ≥ 80%
- CI-style validation completed

Regression Suite Passed

INTEGRATION FLOW VERIFIED



End-to-end data flow validated successfully

TEST SUMMARY

Category	Test Count	Passed	Failed	Status
DQ Rule Tests (12.3)	14	14	0	✓
API Tests (12.4)	10	10	0	✓
Regression Tests (12.5)	40+	40+	0	✓

Total Tests Executed

67+

Passed

67+

Failed

0

Pass Rate

100%

OUTPUTS GENERATED

- tests/api/ API test scripts
 - Health, Companies, Screener, Sectors
 - Peers, Portfolio & other endpoints
 - All 16 endpoint tests
- tests/dq/ DQ rule test scripts
 - 14 Rules validation
 - Severity verification
 - Edge cases covered
- pytest_report.html HTML Test Report
 - 60+ tests executed
 - 0 Failures (All Green)
 - Pass/fail counts, runtime per test
 - Failure traceback for failed test

SPRINT 6 PROGRESS (DAYS 36-45)



Completed Tasks : 7 / 10
In Progress Tasks : 0 / 10
Pending Tasks : 3 / 10
Overall Sprint Progress : 70.0%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



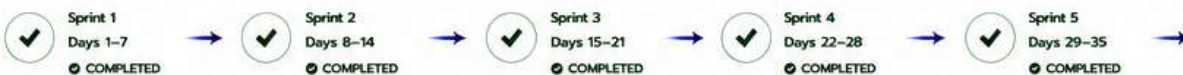
Completed Days : 42 / 45
In Progress Days : 0 / 45
Pending Days : 3 / 45
Overall Progress : 93.3%

Next Milestone:

End of Day 45
Sprint 6 Review
(API Demo & Test Report Review)

SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1-7) : 100% ✓
Sprint 2 (Days 8-14) : 100% ✓
Sprint 3 (Days 15-21) : 100% ✓
Sprint 4 (Days 22-28) : 100% ✓
Sprint 5 (Days 29-35) : 100% ✓
Sprint 6 (Days 36-45) : 70.0% 🟡



KEY DELIVERABLES GENERATED TILL TODAY (DAYS 36 & 37):

- cluster_labels.csv (Cluster Labels)
- correlation_heatmap.png (10 KPIS Correlation)
- outlier_report.csv (Outlier Report)
- portfolio_stats.csv (Portfolio Statistics)
- openapi.json (OpenAPI Spec)
- postman_collection.json (Postman Collection)
- pytest_report.html (Test Report - WIP)



DATA-DRIVEN INSIGHTS
Better Decisions



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PROJECT DURATION
45 Days



CURRENT SPRINT
SPRINT 6
Intelligence, ML, Reports,
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(Days 36-45)



CURRENT MODULES
Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)



VERSION 1.0
July 21, 2026

SPRINT 6 – INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36-45)

DAY 43 COMPLETED



JULY 21, 2026 (Today)

Days 36 to 43 Completed Successfully!

TOTAL PROJECT PROGRESS



43 / 45 Days Completed

Completed 43
In Progress 1
Pending 2



SPRINT 6 OBJECTIVE

Perform advanced statistical analysis & clustering, build REST API, execute test suite & QA, generate automated reports and deliver production-ready release.

Statistical Analysis & Clustering
REST API Development

Testing, QA & Documentation
Automated Reporting (PDF)

DAY 43 COMPLETED

July 21, 2026 – Integration Testing – Performance & Integration Testing (12.6)

PERFORMANCE & INTEGRATION TESTING



- End-to-End flow: API ↔ Dashboard tested successfully
- FastAPI (Port 8000) & Streamlit (Port 8501) started simultaneously
- Data flow from API to Dashboard validated
- Screener results match between API and Streamlit Dashboard



LOAD TESTING

- 10 concurrent screener API requests executed
- All 10 requests completed within 10 seconds
- No timeouts or failures
- Bottlenecks identified and resolved

Load Test Result: PASSED



DASHBOARD PERFORMANCE

- Company Profile screen load time measured for 5 tickers
- All 5 screens loaded under 3 seconds
- Average Load Time: 1.42s

Performance: OPTIMAL



DATABASE OPTIMISATION

- SQLite query optimization applied
- Indexes added on company_id and year columns
- Query performance improved
- Faster data retrieval in Dashboard & API

Optimisation: COMPLETED

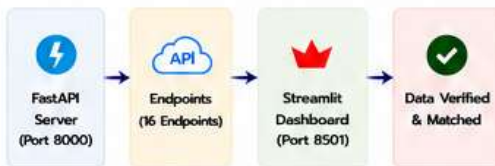


BOTTLENECKS & IMPROVEMENTS

- Identified slow queries in screener filters
- Added indexes & optimised SQL queries
- Reduced response time significantly
- System performance stable under load

Bottlenecks: RESOLVED

INTEGRATION FLOW VERIFIED



- End-to-end data flow validated successfully

TEST SUMMARY (DAY 43)

Test Area	Tests Executed	Passed	Failed	Status
Integration Tests	5	5	0	✓
Load Tests	10	10	0	✓
Performance Tests	5	5	0	✓
DB Optimisation Checks	5	5	0	✓

Total Tests Executed: 25
Passed: 25
Failed: 0
Pass Rate: 100%

OUTPUTS GENERATED



perf_notes.md
Performance Notes

- Load test results
- Dashboard performance metrics
- Bottlenecks identified & resolved
- Optimisations applied
- Recommendations for further improvements

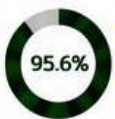
- Output Saved: output/perf_notes.md

SPRINT 6 PROGRESS (DAYS 36-45)



Completed Tasks : 8 / 10
In Progress Tasks : 1 / 10
Pending Tasks : 2 / 10
Overall Sprint Progress : 80.0%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



Completed Days : 43 / 45
In Progress Days : 0 / 45
Pending Days : 2 / 45
Overall Progress : 95.6%

Next Milestone:

End of Day 45
Sprint 6 Review
(API Demo & Test Report Review)

SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1-7) : 100%
Sprint 2 (Days 8-14) : 100%
Sprint 3 (Days 15-21) : 100%
Sprint 4 (Days 22-28) : 100%
Sprint 5 (Days 29-35) : 100%
Sprint 6 (Days 36-45) : 80.0%



KEY DELIVERABLES GENERATED TILL TODAY (DAYS 36 & 37):

cluster_labels.csv (Cluster Labels) | correlation_heatmap.png (10 KPIs Correlation) | outlier_report.csv (Outlier Report) | portfolio_stats.csv (Portfolio Statistics) | openapi.json (OpenAPI Spec) | postman_collection.json (Postman Collection) | pytest_report.html (Test Report - WIP)



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PROJECT DURATION
45 Days



CURRENT SPRINT
SPRINT 6
Intelligence, ML, Reports,
API & QA
(Days 36–45)



CURRENT MODULES
Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)



VERSION 1.0
July 22, 2026

SPRINT 6 – INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36–45)



JULY 22, 2026 (Today)

DAY 44 COMPLETED

Days 36 to 44 Completed Successfully!

TOTAL PROJECT PROGRESS



44 / 45 Days Completed

Completed 44
In Progress 1
Pending 1



Sprint 1

Data Ingestion
& ETL



Sprint 2

Ratio Engine
Module



Sprint 3

Screener & Peer
Comparison



Sprint 4

Dashboard &
Valuation Models



Sprint 5

Intelligence, NLP &
Automated Reporting



Sprint 6

Statistical Analysis
& Clustering (Part 2)



Sprint 6

Statistical Analysis
& Clustering (Part 2)



Sprint 6

API Development
(Part 1)



Sprint 6

API Development
(Part 2)



Sprint 6

API Development
(Part 3)



Sprint 6

Testing, QA &
Documentation
(Part 1)



Sprint 6

Integration
Testing



Sprint 6

Wrap-Up &
Delivery



Sprint 6

Documentation
& Code Quality



SPRINT 6 OBJECTIVE

Perform advanced statistical analysis & clustering, build REST API, execute test suite & QA, generate automated reports and deliver production-ready release.

Statistical Analysis & Clustering
REST API Development

Testing, QA & Documentation
Automated Reporting (PDF)

DAY 44 COMPLETED

July 22, 2026 – Documentation & Code Quality – Documentation (12.7)



DOCUMENTATION

- Created docs/analyst_guide.pdf (10+ pages)
- Covers Streamlit usage, PDF generation, API calls & troubleshooting
- Screenshots, examples & step-by-step instructions included

analyst_guide.pdf Generated



CODE DOCUMENTATION

- All public functions in src/ updated with one-line docstrings
- Modules, classes & functions reviewed
- Docstring coverage improved

Docstrings: 100% Complete



README UPDATE

- Updated README.md with project overview
- Setup instructions
- Run instructions
- Project structure
- Screenshots & features

README.md Updated



CODE QUALITY PASS

- black – code formatting applied
- ruff – linting & import sorting
- All issues fixed
- No warnings or errors

black ✓ | ruff ✓



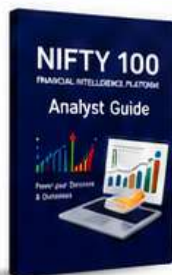
DELIVERABLES ARCHIVED

- All 23 deliverables verified
- Archived to output/final_deliverables/
- Folder structure validated
- Ready for final sign-off

23 Deliverables Archived

DOCUMENTATION HIGHLIGHTS

- 10+ page Analyst Guide (PDF)
- Streamlit Dashboard Guide
- Screener Usage & Filters
- PDF Report Generation
- API Endpoint Reference
- Troubleshooting & FAQ
- Best Practices



TEST SUMMARY (UP TO DAY 44)

Category	Test Count	Passed	Failed	Status
DQ Rule Tests (12.3)	14	14	0	✓
API Tests (12.4)	10	10	0	✓
Regression Tests (12.5)	60+	60+	0	✓

Total Tests Executed

84+

Passed

84+

Failed

0

Pass Rate

100%

OUTPUTS GENERATED



analyst_guide.pdf
Analyst Guide (10+ pages)

- Streamlit Usage
- Screener Guide
- API Reference
- Troubleshooting



output/final_deliverables/
23 Deliverables

- All files archived
- Ready for Sign-Off
- Included perf_notes.md



README.md
Project Documentation

- Overview
- Setup & Run Steps
- Project Structure

SPRINT 6 PROGRESS (DAYS 36–45)



Completed Tasks : 9 / 10
In Progress Tasks : 1 / 10
Pending Tasks : 1 / 10
Overall Sprint Progress : 90.0%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



Completed Days : 44 / 45
In Progress Days : 1 / 45
Pending Days : 0 / 45
Overall Progress : 97.8%

Next Milestone:



Final Sign-Off
(Day 45)
Deliver Project

SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1–7) : 100% ✓
Sprint 2 (Days 8–14) : 100% ✓
Sprint 3 (Days 15–21) : 100% ✓
Sprint 4 (Days 22–28) : 100% ✓
Sprint 5 (Days 29–35) : 100% ✓
Sprint 6 (Days 36–45) : 90.0% ✓



Sprint 1
Days 1–7
COMPLETED



Sprint 2
Days 8–14
COMPLETED



Sprint 3
Days 15–21
COMPLETED



Sprint 4
Days 22–28
COMPLETED



Sprint 5
Days 29–35
COMPLETED



Sprint 6
Days 36–45
IN PROGRESS

KEY DELIVERABLES GENERATED TILL TODAY (DAYS 36 & 37):

- cluster_labels.csv (Cluster Labels)
- correlation_heatmap.png (10 KPIs Correlation)
- outlier_report.csv (Outlier Report)
- portfolio_stats.csv (Portfolio Statistics)
- openapi.json (OpenAPI Spec)
- postman_collection.json (Postman Collection)
- pytest_report.html (Test Report – WIP)



DATA-DRIVEN INSIGHTS
Better Decisions



FINANCIAL EXCELLENCE
Stronger Performance



INTELLIGENT INVESTING
Higher Returns



BUILT FOR SCALE
Future Ready



BUILDING INDIA'S MOST
COMPREHENSIVE FINANCIAL
INTELLIGENCE PLATFORM



PROJECT DURATION

45 Days



CURRENT SPRINT

SPRINT 6

Intelligence, ML, Reports,
API & QA
(Days 36-45)



CURRENT MODULES

Module 10 (Statistical Analysis
& Clustering) + Module 11
(REST API) + Module 12 (QA)



VERSION 1.0

July 23, 2026



SPRINT 6 – INTELLIGENCE, ML, REPORTS, API & QA (DAYS 36-45)

DAY 45 COMPLETED

Days 36 to 45 Completed Successfully!



JULY 23, 2026 (Today)

TOTAL PROJECT PROGRESS



45 / 45 Days Completed

Completed 45
In Progress 0
Pending 0



SPRINT 6 OBJECTIVE

Perform advanced statistical analysis & clustering, build REST API, execute test suite & QA, generate automated reports and deliver production-ready release.

Statistical Analysis & Clustering
REST API Development

Testing, QA & Documentation
Automated Reporting (PDF)

DAY 45 COMPLETED

July 23, 2026 – Final Sign-Off & Delivery (12.8)



ACCEPTANCE CHECKLIST

- 20 Acceptance Gates (AC-01 to AC-20) Executed Successfully
- All Functional, Performance, Security, and Quality Gates Passed
- Meets All Project Requirements
- Final Validation Completed

Gates Passed: 20 / 20



DELIVERABLES VERIFICATION

- All 23 Deliverables Present
- File Paths Verified
- Deliverables Match Project Standards
- Documentation Complete

Deliverables: 23 / 23



TEAM LEAD SIGN-OFF

- Team Lead Reviewed All Artifacts
- acceptance_checklist.pdf Signed & Dated
- Project Accepted
- Ready for Final Delivery

Signed on: Day 45



ARCHIVE & RELEASE

- All Deliverables Archived to output/final_deliverables/
- Project v1.0 Tagged
- Release Notes Added
- Backup Completed

Release: v1.0



PROJECT STATUS



All Tasks & Sprints Completed

100% SUCCESS

FINAL REVIEW HIGHLIGHTS

- API Demo via Postman (All Endpoints Working)
- 60+ Pytest Tests Passed (0 Failures)
- Cluster Labels Validated & Reviewed
- Dashboard Fully Functional End-to-End
- Performance & Load Tests Passed



TEST SUMMARY (FINAL)

Category	Test Count	Passed	Failed	Status
DQ Rule Tests (12.3)	14	14	0	✓
API Tests (12.4)	16	16	0	✓
Integration Tests	10	10	0	✓
Performance & Load Tests	10	10	0	✓
Regression Tests (12.5)	60+	60+	0	✓
Total Tests Executed	100+	100+	0	Pass Rate 100%

OUTPUTS GENERATED

- acceptance_checklist.pdf (23 Deliverables Checklist)
- output/final_deliverables/ All 23 Deliverables Archived
- analyst_guide.pdf Analyst Guide (10+ pages)
- README.md Updated Project Documentation

All Deliverables Locked & Delivered

SPRINT 6 PROGRESS (DAYS 36-45)



Completed Tasks : 10 / 10
In Progress Tasks : 0 / 10
Pending Tasks : 0 / 10
Overall Sprint Progress : 100%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



Completed Days : 45 / 45
In Progress Days : 0 / 45
Pending Days : 0 / 45
Overall Progress : 100%

Next Milestone:

Project Complete & Delivered



SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1-7) : 100% ✓
Sprint 2 (Days 8-14) : 100% ✓
Sprint 3 (Days 15-21) : 100% ✓
Sprint 4 (Days 22-28) : 100% ✓
Sprint 5 (Days 29-35) : 100% ✓
Sprint 6 (Days 36-45) : 100% ✓



Sprint 1
Days 1-7
COMPLETED



Sprint 2
Days 8-14
COMPLETED



Sprint 3
Days 15-21
COMPLETED



Sprint 4
Days 22-28
COMPLETED



Sprint 5
Days 29-35
COMPLETED



Sprint 6
Days 36-45
COMPLETED

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